

UNIT 5

ENVIRONMENTAL CHANGES

Learning objectives:

- Explain how living things can change their environment.
- Identify how people impact the environment positively and negatively.
- Describe how weather and natural events can change environments.
- Explain how changes in the environment affect plants and animals.
- Understand the importance of reducing, reusing, and recycling.
- Define and identify pollution and its effects on the environment.
- Recognize how living things depend on each other in an environment.
- Explain how new plants and animals can change an environment.
- Understand the meaning of extinction and how it happens.
- Describe how fossils help us learn about living things from the past.

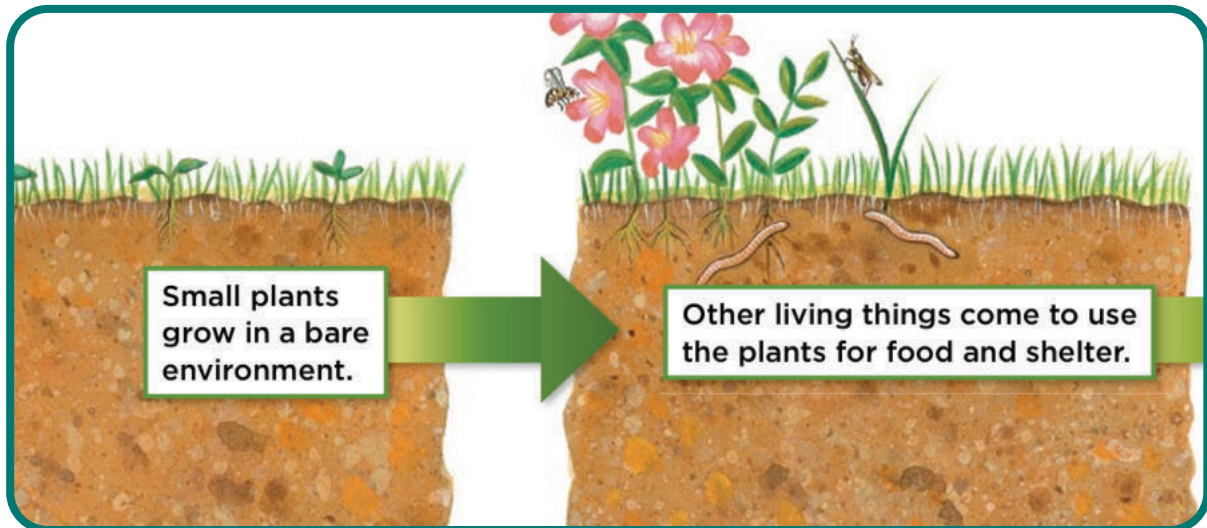


**How do changes to environments
affect living things?**

5.1 LIVING THINGS CHANGE THEIR ENVIRONMENT

How do living things change their environment?

Living things make changes to their environments. Some changes are small. A bird might make a nest. Other changes can be large. Living things might break down a fallen tree. This can make the ground a better place for new plants.



Plants and animals compete with one another to get what they need. They all try to get food, water, sunlight, and shelter in the same place. If some cannot get what they need, they will die. Environments can change as living things compete to meet their needs.

Quick Check

What might happen to a plant if its sunlight is blocked?



How does a beaver change its environment?

Beavers make many changes to their environment. They use their strong teeth to cut down trees and plants. They build dams across streams. This stops the water from flowing and makes a pool or pond. The beavers then build homes in the water.



Beavers use fallen trees, branches, and mud to build their dams and homes.



Quick Check

What adaptation of beavers helps them to take down trees?

Beaver dams cause big changes to the environment. Dry land behind the dam floods with water. The dam might stop water from flowing to other places. Plants may die, and other animals may have to move to get what they need. Some changes might be good. Fallen trees might make room for more plants to grow. Wetlands may form and attract new living things.

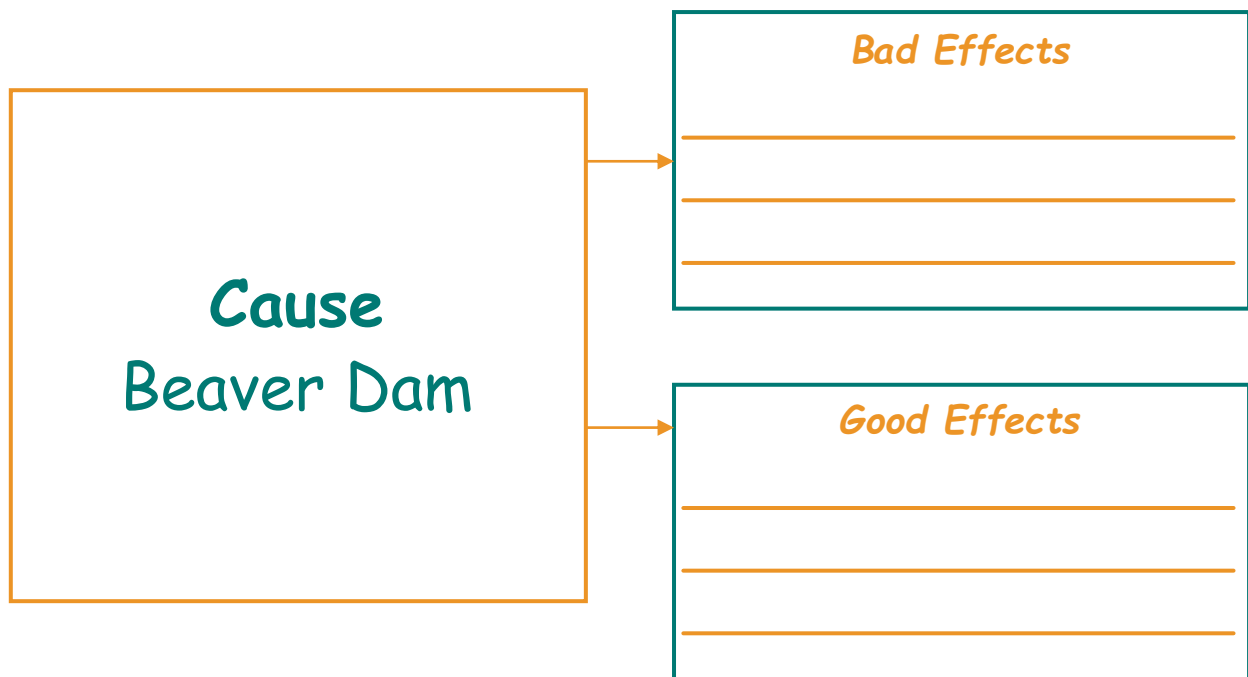




↑ A beaver dam can turn a running stream into a deep pool of water.

Quick Check

Use the diagram below to tell how a beaver dam can cause both good and bad effects.



How do people change their environment?

People make more changes to the environment than any other living thing. Some changes can be good. Others can be harmful. People make a lot of trash. Some of this trash can get into the environment. This is called **pollution**. It can harm the living things of the land, water, and air.



Quick Check

Tell whether each statement is true or false

1. *Most trash in the United States is burned.* _____
2. *Pollution can hurt living things in the water.* _____

People can practise these "**3 Rs**" to cut back on trash and pollution.

- **reduce**: use less of something
- **reuse**: use things more than once
- **recycle**: make new things out of old things



These acts will help us keep our environment healthy.



↑ The things in this recycling bin can be used to make other products.

Quick Check

Write about a way in which people can help the environment.

5.2 CHANGES AFFECT LIVING THINGS

What are some ways environments change?

Weather can change an environment. Rain can make plants grow and blossom. Too much rain can cause floods. Not enough rain can turn green plants brown. Lightning can start fires in a dry grassland or forest. Some changes last only a short time. Some last much longer. Earthquakes, storms, and volcanic eruptions can make sudden changes. These changes might last for years.





↑ Too much rain caused a flood in the this field.

Some living things recover quickly from environmental changes. Grassland grasses might come back soon after a fire. But trees may take many years to grow back. Some changes can last forever. People may cut down all the trees in a place to build roads. Pollution in rivers and lakes can be lasting.



↑ Plants start to grow again after a volcanic eruption.





Quick Check

What change in the environment might result from too much rain?

- Too much rain can cause flooding, which may result in;
- Damage to homes, roads and crops.
- Soil erosion.
- Water pollution.

How do changes affect plants and animals?

A **habitat** is a living thing's home. Changes to a habitat can hurt the living things there. Their water might dry up. Food might become hard to find. Their adaptations may not help them survive big changes. Some living things may need to find a new place to live. Others may need to find new ways to live. Those that cannot move or change their ways might die.



This animal depends on water in its habitat.



During a dry spell, water in the savanna might dry up. Some animals would need to move to find new water. Others might dig down into the ground and wait there for rain. Plants cannot move when their habitats change. They might die. Animals such as fish might also die if their habitat dries up.

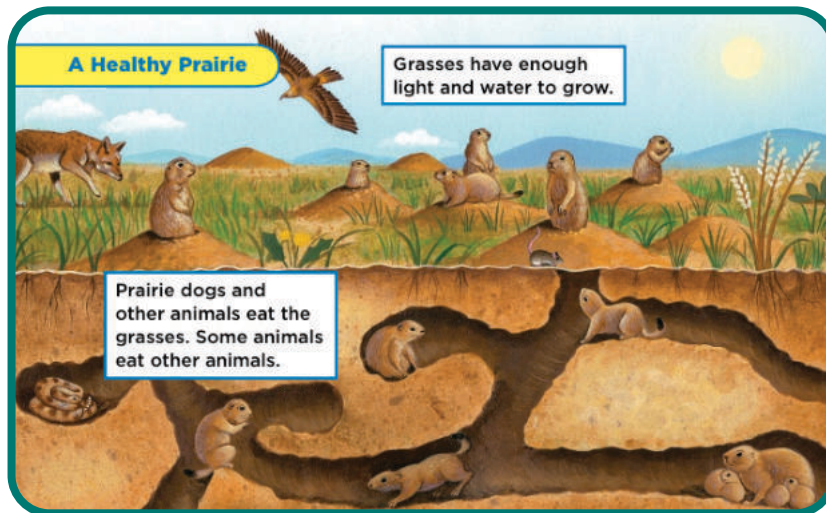


↑ These elephants must move to find food and water when the weather stays dry.

How do living things depend on each other?

Living things may depend on each other for their needs. Many animals use plants for food, water, and shelter. Some animals eat other animals. They may all share the same habitat. Change may happen to one part of the habitat. One living thing might be harmed. That could lead to other changes in the habitat. All the living things in an environment are linked together.





Prairie dogs eat the grasses in their habitat.

They dig burrows to stay safe. Other animals may use these burrows. Big animals eat the smaller animals. If the small animals went away, the whole environment would change.

Quick Check

What might happen to an animal's habitat if no rain fell for many months?



Quick Check

Use the graphic organizer below to list two effects of dry weather on the prairie.

Cause
Dry weather



Effect

What happens when new living things move in?

Living things sometimes move from one environment into another. Animals might move to find new sources of food and water. Seeds from plants might blow on the wind or float down a river. People can also bring new plants and animals into a new place.



↑ People planted a new kind of tree here.
Now the older trees cannot get what they need.



New plants and animals can cause changes in an environment. They will use food and water that other plants and animals need. They might also bring sickness with them. People once raised foxes for their fur. Then they let them go. Now the foxes try to get the same food and water as other animals in the wild.



↑ This fox competes with other animals for food and space.

Quick Check

How might a new kind of illness get from one place to another?

5.3 LIVING THINGS OF THE PAST

What can happen if the environment suddenly changes?

Thousands of years ago, ice covered much of the United States. Then it grew warmer. The ice melted and some plants and animals could not survive in the warmer environment. They became **extinct** over time. Extinct means there are no more left alive.





↑ This woolly mammoth became extinct when its environment grew warmer.

Some plants and animals become extinct

due to changes in weather patterns. Others may die of sickness. The things people do to an environment can make some living things extinct.

Change	Living Thing	What Might Happen	Why
warmer climate	saber-toothed cat	becomes extinct	unable to find food; unable to survive in warm climate
volcanic eruption	short-tailed albatross	survives	flies to new environment
colder climate	bear	survives	grows thicker fur



How can we learn about things that lived long ago?

Fossils are the hardened remains of things that were once alive. They teach us about living things from long ago. Some fossils can show us the size and shape of a plant or animal. Fossils of teeth can tell us what an animal ate. Animals with sharp teeth could eat meat. Animals with flat teeth ate plants. Fossils of bones can tell us how an animal moved. Foot and leg bones show if the animal could run or climb. Wing bones tell that an animal could fly.

This fossil shows an animal →
that had wings like a bird.



Fossils can also tell how Earth has changed.

Some fossils of fish are found on land. This tells us that water covered that land long ago. Layers of rock tell about Earth's history. Fossils found deep in the ground are the remains of the oldest plants and animals.



↑ A fish fossil can tell us
where oceans used to be.



Quick Check

Draw a line between the kind of fossil and what it tells us.

flat teeth

animal ate meat

wing bones

animal ate plants

sharp teeth

animal could fly

How are living things today similar to those that lived long ago?

Some extinct animals are like animals of today. Woolly mammoths were like elephants. They were big and had tusks. They lived in groups. They had flat teeth and ate plants. Mammoths were much bigger than elephants, though. Their tusks were much longer. They had thick hair and lived in cold habitats.



The woolly mammoth and the elephant look similar.



Some dinosaurs of the past looked like lizards today. They had similar body parts. Maybe they used these body parts the same way, too. Fossils can tell us if the animals ate the same foods. They can also tell if the animals moved the same way.

The dilophosaurus dinosaur →
(left) looked like the frilled
lizard of today.



Quick Check

Tell whether each body part belonged to a mammoth, to an elephant, or to both.

flat teeth

thick hair

Have some animals stayed the same over time?

Some animals have adaptations that work well over time. They can survive great changes in their environments. Fossils of such animals look almost the same as the animals today.





↑ The fossil of a crocodile
from long ago



↑ A crocodile today

Shrimp survived when salt levels changed in the oceans. They have been nearly the same for over 100 million years. The cockroach is an example of an insect that has stayed about the same for millions of years.



↑ A cockroach



↑ A shrimp



 Quick Check

Why might you compare a fossil from long ago to a living thing today?

How are shrimp, crocodiles, and cockroaches different from most other living things?

Vocabulary Review

Rearrange the letters to form vocabulary words. Then write the word's number after its definition below.

1. textnic _____

2. esure _____

3. thatiba _____

4. crelecy _____

5. lofsis _____

6. lipolotun _____

7. cerude _____



a. the hardened remains of a living thing _____

b. to use less of something _____

c. harmful things getting into the air, water, or land

d. all dead, no more of that living thing alive on Earth

e. to use something again _____

f. the place where a living thing makes its home

g. to turn an old thing into something new _____



Test Your Understanding

Q1. What is pollution?

Ans: _____

Q2. How do beavers change their environment?

Ans: _____

Q3. What can people do to reduce trash and pollution?

Ans: _____

Q4. How does too much rain affect plants?

Ans: _____

Q5. What happens to animals when their habitat changes?

Ans: _____

Q6. How do plants move into new environments?

Ans: _____



Q7. What is a fossil?

Ans: _____

Q8. What does it mean if an animal is extinct?

Ans: _____

Q9. How do fossils help us learn about animals from long ago?

Ans: _____

Q10. How can people help keep the environment clean?

Ans: _____



Fill in the blanks.

1. A _____ is when harmful things get into the air, water, or land.
2. To _____ means to use less of something.
3. A _____ is where a living thing makes its home.
4. Beavers build _____ to stop water from flowing.
5. _____ means all of a kind of living thing are dead.
6. _____ are the hardened remains of living things.
7. People can _____ old things to make something new.
8. Living things compete for food, water, sunlight, and _____.
9. When animals move to find food or water, it is called _____.
10. A sudden environmental change, like a fire, can cause _____.



Tick (✓) for True and cross (x) for False statements.

1. Pollution helps plants grow. ()
2. Beavers use mud to build their homes. ()
3. Too much rain can cause floods. ()
4. Fossils tell us about animals that are alive today. ()
5. Living things never change their environment. ()
6. People can recycle to help the environment. ()
7. Extinct animals still live in some parts of the world. ()
8. Plants can move to new places like animals do. ()
9. A beaver's dam can create new wetlands. ()
10. All environmental changes are harmful. ()



Multiple Choice Questions.

1. What helps reduce trash?

- a) Using more plastic
- b) Throwing everything away
- c) Recycling old materials

2. What might happen if too many new animals move into an environment?

- a) Nothing changes
- b) Other animals might not find enough food
- c) The weather will get colder

3. How do fossils help us?

- a) They show what animals looked like long ago
- b) They tell us about future animals
- c) They make plants grow faster

4. What is one way people can help the environment?

- a) Make more trash
- b) Cut down all the trees
- c) Reduce, reuse, and recycle

5. What adaptation helps beavers cut down trees?

- a) Strong legs
- b) Sharp teeth
- c) Long tails



6.What is one effect of building roads in forests?

- a) Trees grow faster
- b) Animals have more places to live
- c) Trees are cut down, and habitats are destroyed

7.How can seeds from plants move to new environments?

- a) By walking
- b) By floating down a river
- c) By swimming

8.Which of the following is an example of reuse?

- a) Throwing away plastic bottles
- b) Using a cloth bag instead of plastic
- c) Burning trash

9.What might happen if too much rain falls in an area?

- a) The plants will grow faster
- b) The area will flood
- c) The weather will become dry

10.Which of the following animals survived with little change over millions of years?

- a) Mammoth
- b) Cockroach
- c) Saber-toothed cat





Recycling Craft Project



ACTIVITY

Recycling Craft Project

Understand the importance of reduce, reuse, and recycle by creating new items from old materials.

Materials Needed:

- Old newspapers, plastic bottles, cardboard, cans, fabric scraps, glue, scissors, and markers.

Instructions:

1.Introduction: Begin with a brief discussion about how recycling helps reduce pollution and saves natural resources.

2.Activity: Divide the class into small groups and give each group recyclable materials. Ask them to create something useful or decorative (e.g., a pencil holder from a can, a plant pot from a plastic bottle, or a newspaper collage).

3.Presentation: Each group presents their creation and explains how it helps the environment by reusing materials.

